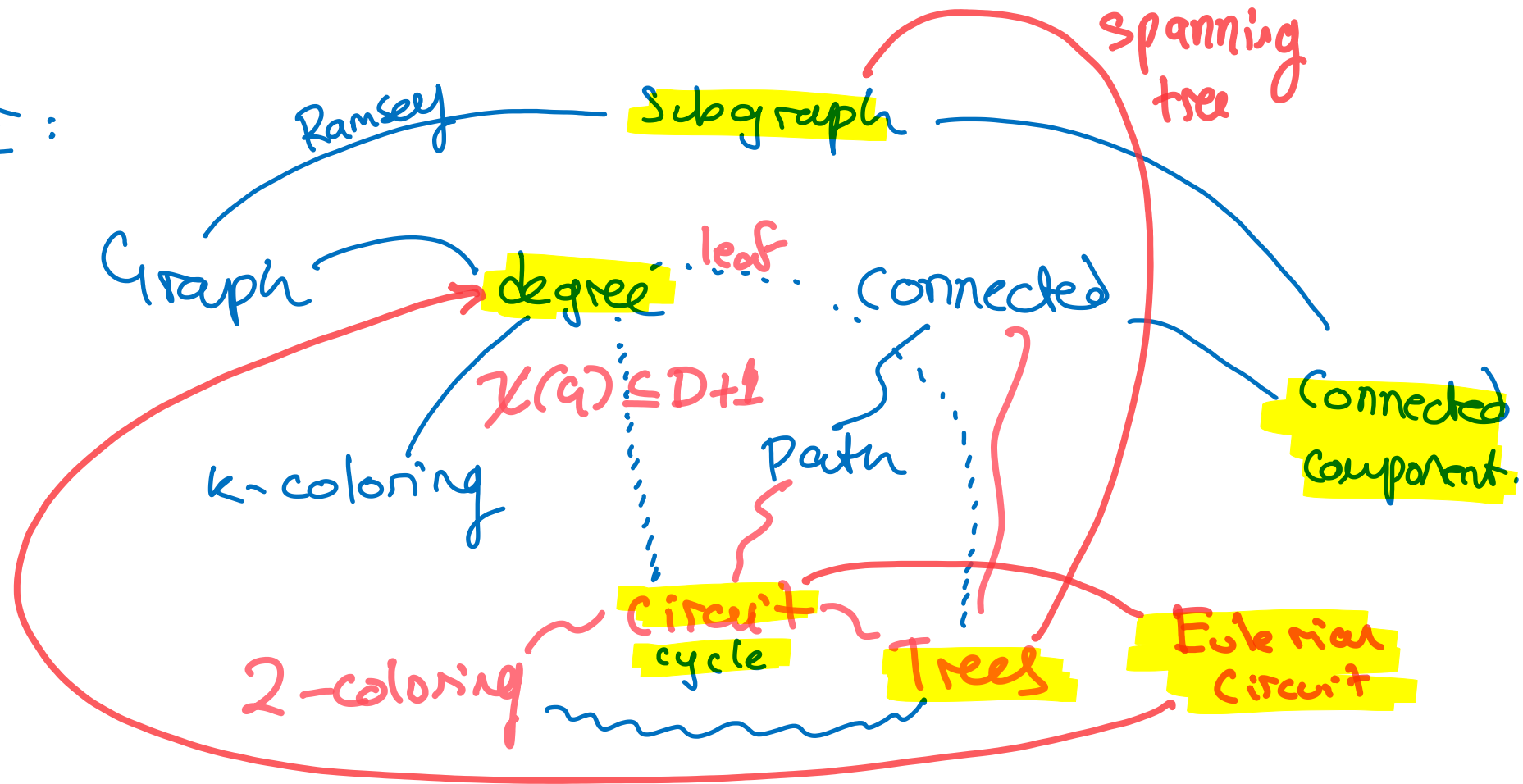


Lecture 17

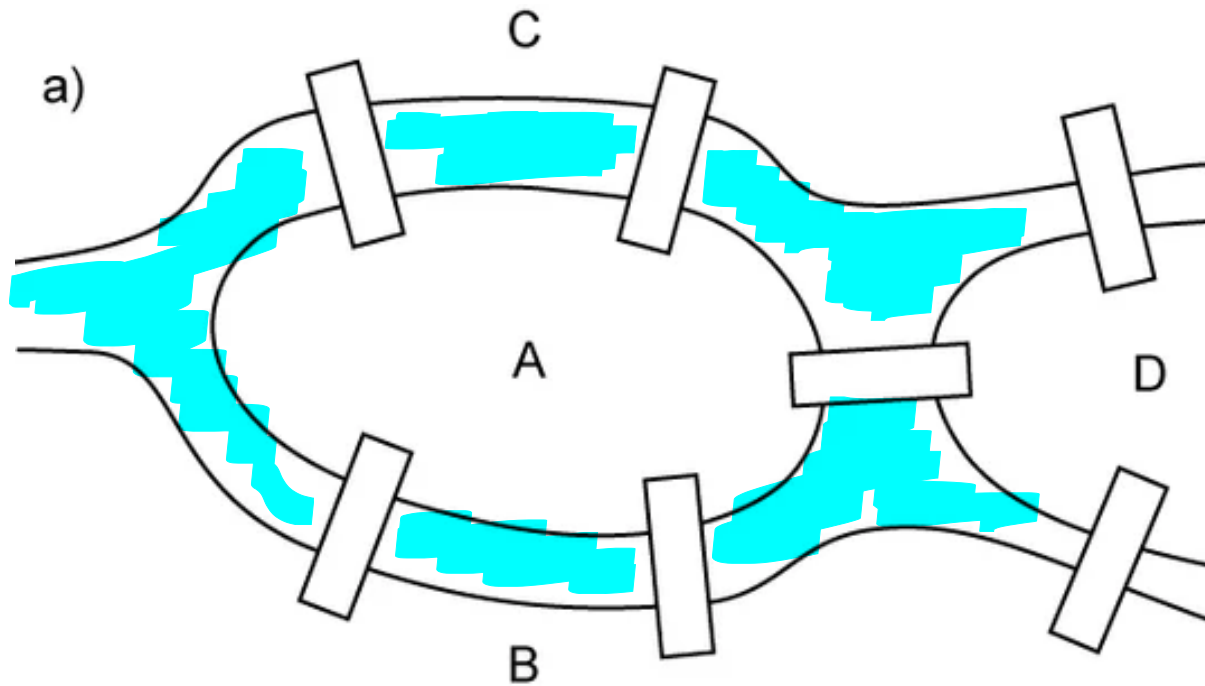
So far:



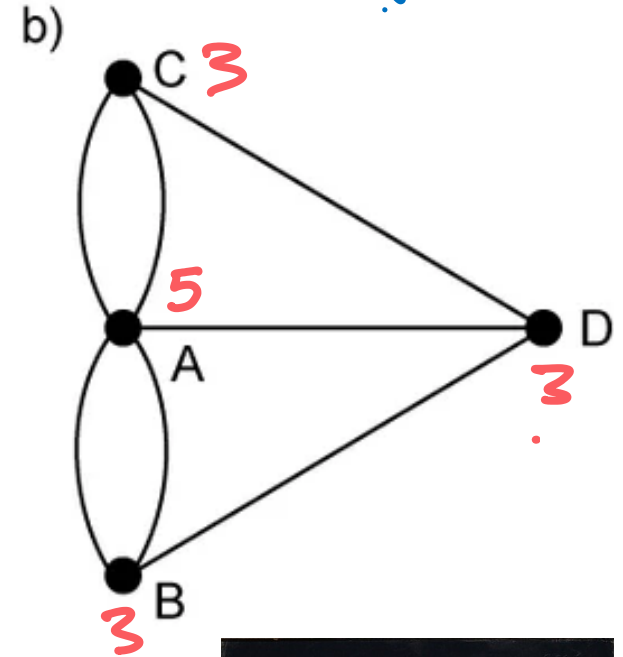
Idea:

- induction on vertices, edges $G-x$, $G-xy$.
- Concatenating, reversing paths.
- Considering connected components
- naming, keywords (traverse, incident, adjacent, ...)

Königsburg 1736:



$G = (V, E)$. Multigraph
 $V = \{A, B, C, D\}$.
 $E = \{\text{bridges}\}$.



Puzzle: Can you walk from A to A traversing every bridge exactly once?

Defn: An Eulerian Circuit in a graph

$G = (V, E)$ is a circuit which traverses every edge $e \in E$ exactly once.



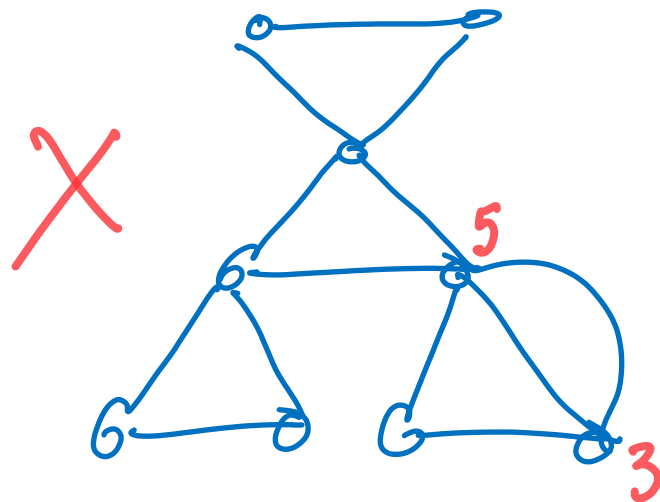
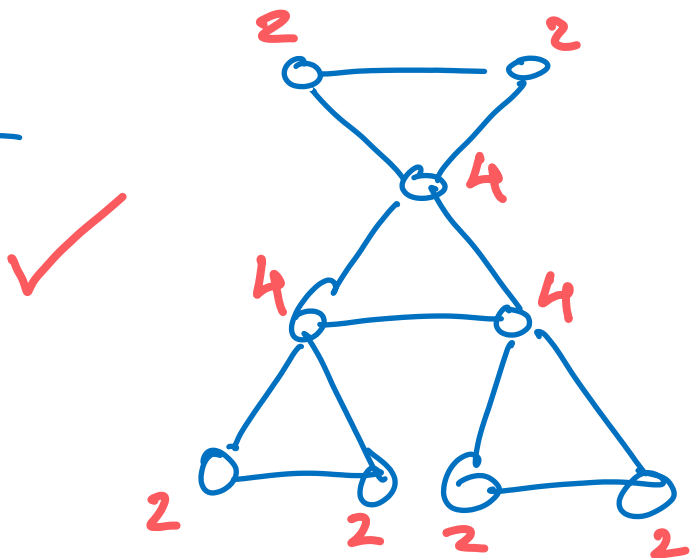
Path
circuit
cycle

Theorem 17.1: If a ^{connected} graph $G=(V,E)$ ^{with $|V| \geq 2$} has an Eulerian circuit
 then $\deg_G(v) \equiv 0 \pmod{2}$ for all $v \in V$.

$[\exists v \deg_G(v) \text{ is odd} \rightarrow G \text{ has no E.C.}]$

Theorem 17.2: If a connected graph $G=(V,E)$ with $|V| \geq 2$
 has $\deg_G(v) \equiv 0 \pmod{2}$ for all $v \in V$, then
 G has an Eulerian circuit.

ex:



Proof of 17-1: Assume $G=(V,E)$ is connected and $|V| \geq 2$.

Assume $C = x_0, e_1, x_1, \dots, e_m, x_m, x_m = x_0$ is an Eulerian ckt of G .

Assume $v \neq x_0$ is a vertex of G .

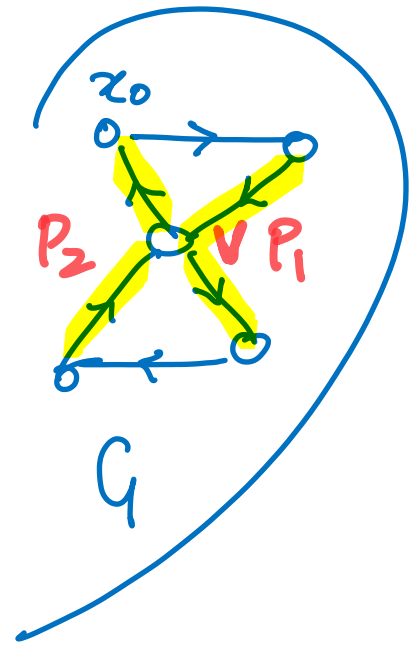
Suppose C visits v exactly k times.

Observe that in each visit, C traverses exactly two distinct edges incident with

v — call these pairs of edges $P_1, P_2, \dots, P_k \subseteq E$.

Moreover, these pairs must be disjoint since C is Eulerian, and every edge incident with v must appear in a pair P_i . Thus,

$$\deg_G(v) = |P_1| + |P_2| + \dots + |P_k| = 2k, \text{ as desired.}$$



Since the number of vertices of odd degree in G is even, $\deg(x_0)$ must be even.

(Alternatively: Consider $C' = x_1, e_2, x_2, \dots, x_m, e_1, x_1$) $\square$

Lemma: If $G=(V,E)$ is a graph with $\deg_G(v) \geq 2$ for all $v \in V$, then G contains a cycle. (as a circuit.)

Proof: By contrapositive. Assume G is acyclic. We will produce a $v \in V$ s.t. $\deg(v) < 2$. Let

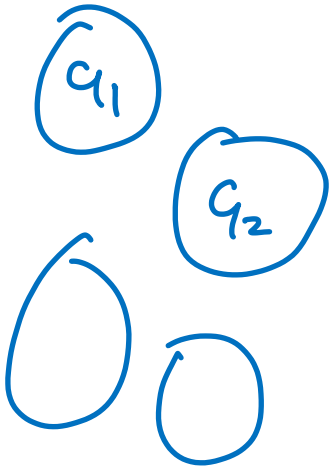
$G = G_1 \cup G_2 \cup \dots \cup G_k$ be the connected components of G .

Since G_i is also acyclic, it is a tree.

If G_i is an isolated vertex v , then

$$\deg_G(v) = \deg_{G_i}(v) = 0$$

Otherwise, G_i has a leaf v , with $\deg_G(v) = \deg_{G_i}(v) = 1$ $\square$



Proof of 17-2: By induction on the number of edges m .

Let $P(m) =$ "If $G=(V,E)$ is connected, $|V| \geq 2$, all even degrees then G has an Eulerian ckt."

We will show $P(m)$ for all $m \geq 2$. $\langle m=1 \text{ is vacuous} \rangle$.

Base Case: Any connected graph with $|V| \geq 2, |E| = 2$

is isomorphic to



not allowed

Of these, only G_2 has even degrees, and G_2 has an Eulerian ckt so $P(2)$ holds.

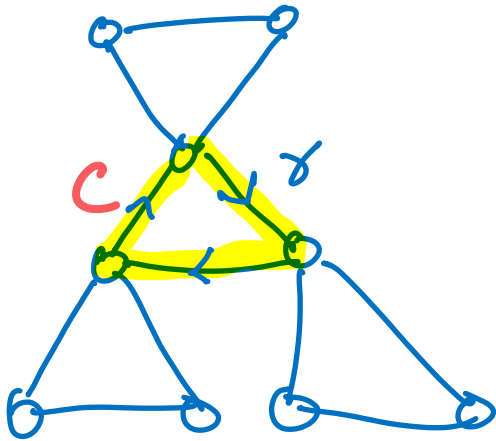
Inductive Step: Assume $m \geq 2$, $P(2), P(3), \dots, P(m)$ hold.

We will show $P(m+1)$. Assume $G=(V,E)$ is connected, $|V| \geq 2$, has all even degrees.

$\langle$ Can't delete 1 vertex or 1 edge — mess up degrees! $\rangle$

Since $\deg_G(v) \geq 2$ for all $v \in V$, the Lemma yields a cycle γ in G . Let

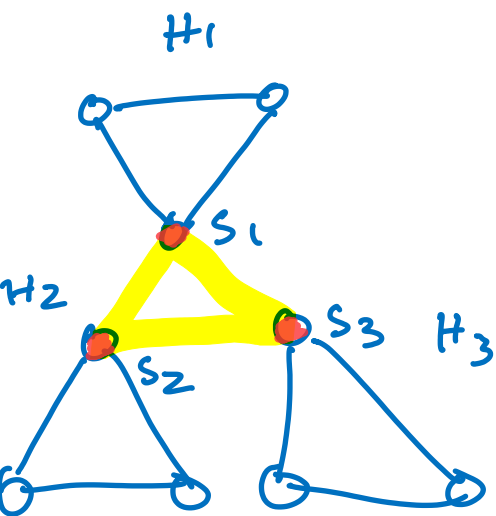
$C = (V_C, E_C)$ be the subgraph traversed by γ .



G

Case 1: $C = G$. In this case γ is an Eulerian ckt of G , since γ is a cycle.

Case 2: $C \neq G$. Let $H = G - C = (V, E - E_C)$ be the graph obtained by removing the edges of C from G .



$$H = G - C$$

Let $H_1, \dots, H_k$ be the connected components of H with at least 2 vertices.

Observe that for $i=1, \dots, k$

① H_i is connected and has ≥ 2 vertices.

② The degree of every vertex $v \in H_i$ is
 $\deg_{H_i}(v) = \deg_G(v) - \deg_C(v).$

Since $\deg_C(v) \in \{0, 2\}$ and $\deg_G(v) \equiv 0 \pmod{2}$,

$$\deg_{H_i}(v) \equiv 0 \pmod{2}.$$

③ If $H_i = (V_{H_i}, E_{H_i})$ then $|E_{H_i}| \leq |E| - |E_C|$

$$\leq |E| - 2$$

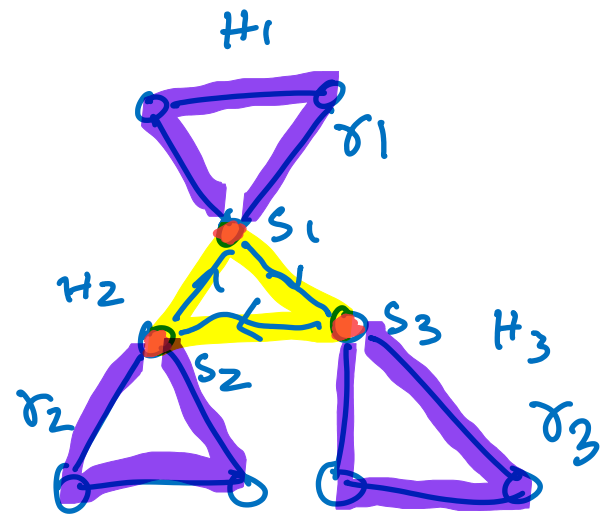
$$\leq m+1-2$$

$$\leq m-1.$$

cycle,
 $\downarrow$
 ≥ 2

Claim: For every $i=1, \dots, k$, there exists a vertex $s_i \in H_i$ such that γ visits s_i . $\langle HW \rangle$

By induction, each H_i for $i=1, \dots, k$ has an Eulerian circuit γ_i . WLOG, we may assume each γ_i starts and ends at s_i .



$$H = G - C$$

$$\gamma = \gamma \dots s_1 \dots s_2 \dots s_3 \dots$$

$\downarrow$
 γ_1

$\downarrow$
 γ_2

$\downarrow$
 γ_3

Let σ be equal to γ with the first occurrence of s_i replaced by γ_i for $i=1, \dots, k$.

- Observe that:
- ① σ is a circuit.
 - ② σ traverses every edge of C exactly once.
 - ③ σ traverses every edge of $H_1 \cup H_2 \cup \dots \cup H_k$ exactly once.

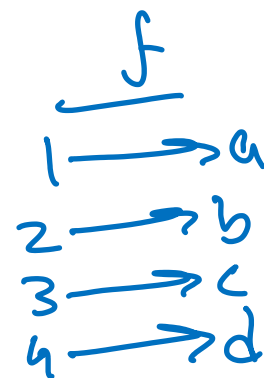
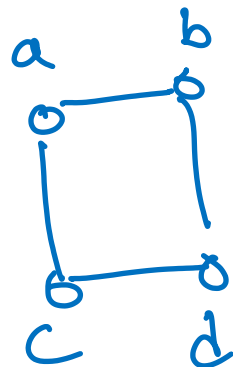
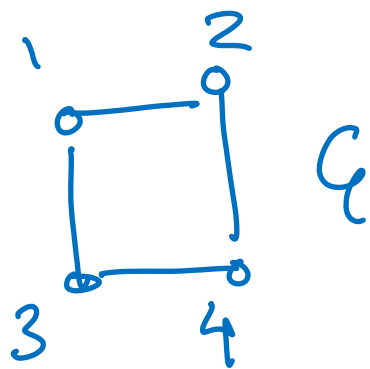
Since $C \cup H_1 \cup H_2 \cup \dots \cup H_k$ contains all the edges of G , σ is an Eulerian ckt, as desired. $\square$

- Remarks:
- Needed S_i to glue together γ_i to get σ .
 - Needed to remove a cycle to preserve even degrees.
 - 3 ways to do induction:
 - ① remove vertex
 - ② remove edge.
 - ③ remove subgraph.

Defn: Two graphs $G = (V, E)$ and $H = (W, F)$ are isomorphic if there is a bijection
 $f: V \rightarrow W$

s.t. $\forall x, y \in V \quad (\{x, y\} \in E \leftrightarrow \{f(x), f(y)\} \in F)$

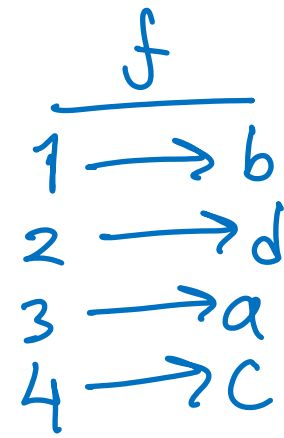
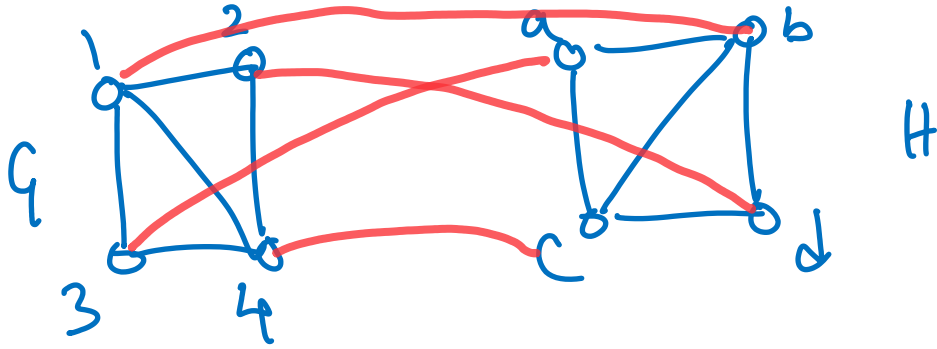
ex:



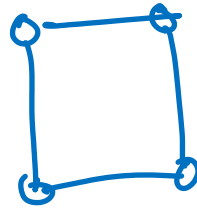
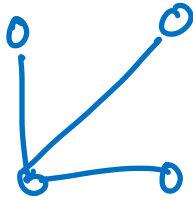
$\deg(4) = 2$. $\hookrightarrow$
 4 is adjacent to 3, 2.

$\deg(f(4)) = 2$
 d is adjacent to c, b

ex:



ex:



not isomorphic.